

CSE 4636: Web Architecture

Project Title: CareerPilot

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Project Overview

CareerPilot is a microservices-driven platform designed to centralize and automate the fragmented job-seeking workflow. Built upon an enriched Spring Boot backend, it replaces manual tracking spreadsheets by isolating computationally heavy ATS resume parsing, live job API scraping, email webhook tracking, and dynamic PDF cover letter compilation into autonomous, high-performance services. This decoupled architecture ensures strict data integrity, fault tolerance, and responsive UI performance while bridging the gap between modern web design and distributed software engineering.

Core Features

- **Live Job Discovery & Skill-Gap Matching:** Aggregates real-time job openings from external public APIs via Spring WebClient, utilizing non-blocking reactive streams to prevent network latency. The backend algorithmically compares fetched job requirements against structured user profile data to calculate a mathematical skill-overlap fit score and automatically surfaces relevant free online courses to bridge identified knowledge gaps.
- **Algorithmic ATS Resume Scorer:** Ingests user-uploaded PDF resumes using Apache Tika document parsing to extract raw text. The backend executes a custom TF-IDF (Term Frequency-Inverse Document Frequency) keyword density algorithm against target job descriptions, stripping stopwords to evaluate semantic relevance, calculate an objective compatibility score, and highlight missing mandatory technical keywords.
- **Smart Email Application Tracker (Auto-Kanban via Webhooks):** Processes simulated email webhook notifications using regular expressions and natural language categorization to automatically detect interview invitations or rejection notices. The system dynamically transitions application statuses across an interactive Kanban board (Wishlist → Applied → Interviewing → Offer) without human intervention and triggers automated reminder alerts via Spring Mail.
- **Dynamic Profile & Automated PDF Cover Letter Generator:** Features a structured profile wizard for entering education, skills, and work history with real-time markdown previews. Paired with an integrated backend document compiler utilizing OpenPDF, the system auto-generates customized, professionally styled, and downloadable PDF cover letters directly on the server.

Technologies to be Used

- **Frontend:** React.js, Vite, Tailwind CSS, shadcn/ui
- **Backend Framework:** Java 21, Spring Boot 3.5, Spring Cloud Gateway
- **Security & Infrastructure:** Spring Security + Stateless JWT, Docker Compose
- **Database:** PostgreSQL
- **Key Backend Libraries:** Apache Tika (Resume Parsing), OpenPDF / Apache PDFBox (Document Generation), Spring WebClient (Reactive API Scraping), Spring Mail (Notifications)